

FIT&FUEL

The Fitness Tracker

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Abstract: The Fit&Fuel application is a comprehensive Flutter-based fitness and nutrition management platform designed to address the fragmented landscape of mobile health and wellness solutions. Conventional fitness applications either focus exclusively on workout guidance or nutrition tracking, compelling users to manage multiple disparate tools for a holistic health management experience. Fit&Fuel bridges this gap by delivering a unified, offline-first platform that integrates structured exercise video tutorials, intelligent calorie and macronutrient calculation, context-aware recipe suggestions, and curated workout playlists within a single cohesive application. Developed using Flutter with the GetX state management framework, the application adopts a feature-driven modular architecture that ensures scalability, maintainability, and high performance across Android devices. The nutrition module employs the Harris-Benedict Basal Metabolic Rate formula to compute personalised daily calorie and macronutrient targets based on user-defined fitness goals, weight gain, loss, or maintenance, while an intelligent error-scoring algorithm matches users with nutritionally compatible recipes from a bundled offline database. The exercise module provides multi-level filtering by body part, muscle group, and equipment type, accompanied by embedded video demonstrations for proper form guidance. An integrated music player supports background audio playback with persistent notification controls, enhancing the workout experience. By consolidating these capabilities into a fully offline platform, Fit&Fuel empowers fitness enthusiasts with comprehensive health management tools accessible without any network dependency.

Keywords: Fit&Fuel, Nutrition Tracking, Calorie Calculator, GetX State Management, Offline-First Architecture, Exercise Tutorial System.

1. Introduction:

The proliferation of smartphones has fundamentally transformed how individuals engage with health and wellness. Mobile fitness applications have emerged as indispensable tools that bridge the gap between professional guidance and everyday personal practice. However, the existing ecosystem of fitness applications is predominantly fragmented: users who seek both structured workout guidance and detailed nutritional management are typically required to maintain multiple applications simultaneously, resulting in a disjointed and inefficient user experience. Moreover, many commercially available fitness platforms impose a dependency on continuous internet connectivity, rendering them inaccessible in environments with limited or unavailable network coverage.

Physical inactivity and poor dietary habits remain the leading contributors to non-communicable diseases globally, including obesity, type 2 diabetes, and cardiovascular conditions. The World Health Organisation reports that insufficient physical activity is among the top risk factors for global mortality. Addressing this challenge requires accessible, personalised, and technologically robust solutions that motivate consistent engagement with fitness and nutrition management. Mobile applications, by virtue of their ubiquity and interactivity, present an ideal medium for such interventions.

Fit&Fuel is developed in response to these challenges. The application leverages Flutter's cross-platform capabilities to deliver a consistent, high-performance experience on Android devices. Its offline-first architecture ensures full functionality without network dependency, making it accessible to a broad spectrum of users irrespective of connectivity constraints. The application consolidates four core domains, exercise, nutrition, recipe management, and motivational audio, within a unified, cohesive interface. Key features include a science-backed calorie and macronutrient calculator employing the Harris-Benedict Basal Metabolic Rate formula, an intelligent macro-based recipe suggestion algorithm, a comprehensive exercise library with embedded video tutorials and multi-level filtering, and a full-featured music player with background playback support. Together, these features position Fit&Fuel as a complete, self-contained health management companion.

This paper presents the design, architecture, and implementation of Fit&Fuel. The subsequent sections provide a literature review of related work, detail the research methodology and technology stack, present the application's key results and feature outcomes, and conclude with observations on contributions and avenues for future enhancement.

2. Literature Review:

Mobile health (mHealth) applications have attracted considerable research interest over the past decade. Klasnja and Pratt examined how smartphones can support health behaviour change, demonstrating that well-designed mobile interventions can yield sustained improvements in physical activity and dietary adherence. Their work underscores the importance of personalisation, accessibility, and motivational scaffolding, principles that directly inform Fit&Fuel's design philosophy.

Nutrition tracking applications such as MyFitnessPal and Cronometer have demonstrated the efficacy of calorie and macronutrient logging in supporting weight management objectives. It is found that consistent self-monitoring via mobile applications significantly correlated with caloric deficit adherence and sustained weight loss. However, these platforms rely on expansive cloud databases and user-generated entries, introducing dependency on network connectivity and raising concerns about data accuracy. Fit&Fuel addresses this limitation through a curated offline recipe and nutrition database with validated macronutrient profiles.

In the domain of exercise guidance applications, research by Bort-Roig et al. highlighted that the provision of instructional video content within fitness applications significantly improves exercise form compliance and reduces injury risk. Fit&Fuel

incorporates embedded video demonstrations for each exercise, with multi-level filtering by body part, muscle group, and equipment type, enabling users to access targeted, contextualised guidance efficiently.

Existing fitness applications that partially overlap with Fit&Fuel's feature set include MuscleWiki exercise tutorials. Neither platform provides an offline-first, unified solution combining exercise demonstration, calorie calculation, recipe matching, and workout audio in a single application. This research gap motivates Fit&Fuel's development as a comprehensive, self-contained fitness management platform.

3. Research Methodology:

The research methodology for the Fit&Fuel project followed a structured, agile-inspired approach. It emphasised a seamless blend of innovative technology and user-centred design to create a comprehensive fitness and nutrition management application. The methodology focused on delivering real-time data accuracy, an intuitive user interface, and complete offline functionality. Key stages of development were grounded in iterative principles, allowing continuous improvements based on ongoing testing and user feedback. Below is a breakdown of the core technologies used in this project and their roles:

- **Flutter (Front-end Development):** Used to build the mobile interface, ensuring the application works smoothly on Android devices with a single Dart codebase. Flutter's widget-based rendering pipeline enables sophisticated, performant UI components, including animated page transitions, expandable cards, and real-time progress indicators.
- **GetX (State Management & Navigation):** Integrated as the primary state management, dependency injection, and navigation framework. GetX's reactive observables (RxList, RxBool) enable automatic UI updates without manual setState calls, while its lazy-put registration pattern ensures optimal resource management and automatic controller disposal upon route exit.
- **JSON Asset Database:** All application data, exercises, recipes, quotes, playlists, and songs are stored as structured JSON files within the application's asset bundle. This eliminates network dependency and guarantees data availability regardless of connectivity status.
- **just_audio & just_audio_background:** Employed for high-fidelity audio playback and background music support. The just_audio_background package enables persistent Android notification controls, allowing users to manage playback without returning to the application foreground.
- **video_player & video_thumbnail:** Used to render embedded exercise tutorial videos and generate preview thumbnails dynamically. A lazy materialisation strategy extracts video assets to temporary storage on demand, minimising initial load time and memory footprint.
- **Harris-Benedict BMR Formula:** The nutrition calculation module implements the Harris-Benedict Basal Metabolic Rate equation, adjusted with a sedentary activity factor (1.2), to derive personalised Total Daily Energy Expenditure (TDEE). Macronutrient targets are calculated from TDEE using goal-specific ratios (gain, lose, or maintain), with protein fixed at 1.8 g per kilogram of body weight.
- **Macro-Based Recipe Suggestion Algorithm:** An error-scoring algorithm computes a weighted composite score across calorie, protein, fat, and carbohydrate deviations for each recipe relative to the user's computed targets. Recipes falling within a configurable threshold (0.30 per-meal, 0.35 total-day) are surfaced as suggestions, enabling nutritionally aligned meal planning.
- **Manually Created Models and Controllers:** Custom-built model classes with json_serializable code generation manage structured de-serialization of all JSON data sources. Dedicated service classes encapsulate asset-loading logic, maintaining a clean separation between the data access and presentation layers.
- **Dart Language:** Powers the logic and structure of the application, enabling efficient handling of asynchronous tasks such as asset loading, video thumbnail generation, and audio stream management through Dart's async/await and Stream primitives.

By leveraging these technologies, the Fit&Fuel application delivers a comprehensive solution for offline fitness and nutrition management, ensuring an engaging, efficient, and reliable user experience across all supported devices.

4. Results:

The Fit&Fuel project achieved its objective of delivering a unified fitness and nutrition management platform successfully. The application enables users to effortlessly manage their exercise routines and dietary goals, leveraging features such as structured video tutorials, personalised calorie calculations, intelligent recipe suggestions, and curated workout playlists. Below are the key outcomes from the user interface design and functional evaluation.

[1] Home Screen: The Home Screen presents a streamlined, motivational entry point featuring a randomly selected daily motivational quote drawn from a curated offline database of over one hundred entries. Below the quote, six colour-coded exercise category cards, representing Chest, Back, Shoulders, Arms, Legs, and Cardio, provide immediate navigational access to the exercise library. The visual design employs a dark navy background with a gold accent colour, creating a high-contrast interface that aligns with contemporary fitness application aesthetics.

[2] Dashboard Screen: The Dashboard Screen is the central performance hub, storing all data locally in a SQLite database. A Body Measurements card at the top displays weight, height, colour-coded BMI, and secondary measurements for chest, waist, biceps, and thighs, with an Update button to record new entries. A row of three stat cards summarises the current week: session count, estimated kilocalories burned, and a consecutive-day streak that tolerates rest days. Below, a scrollable workout history lists session cards showing date, targeted body parts, exercise count, duration, and calories, with tap-to-edit and swipe-to-delete interactions. New sessions are logged via a modal sheet where exercises and sets are composed individually, with calorie expenditure estimated automatically using MET-based formulae scaled to the user's body weight.

[3] Exercise Module: The Exercise Module provides access to a library of over three hundred exercises, organised by body part category selected from the Home Screen. Within each category, users may apply secondary filters by specific muscle group and equipment type, enabling precise drill-down to relevant exercises. Each exercise entry displays a thumbnail preview, name, target muscle group, and required equipment, with an expandable section providing detailed execution instructions. Tapping the

thumbnail activates an inline video player that streams the embedded tutorial directly from the application's asset bundle, without network requests.

[4] Recipe Library: The Recipe Library presents a searchable collection of recipes with real-time search filtering. Each recipe card displays the dish name, calorie content, macronutrient summary, and serving count. Tapping a recipe navigates to a detail view containing a complete ingredient list, step-by-step preparation instructions, and preparation tips. All recipe data is stored locally within the asset bundle, ensuring instant access without latency.

[5] Music Player: The Music Player feature organises curated playlists, including Gym, Motivation, and Cardio collections, containing songs in high-efficiency Opus audio format. Playlists can be expanded to reveal individual track listings. The integrated audio player supports play/pause, next/previous track navigation, and real-time playback progress visualisation via a custom progress bar. Background playback is enabled through the `just_audio_background` integration, maintaining an Android notification control panel that persists across application foreground and background states. A persistent mini-player widget is anchored above the bottom navigation bar, providing continuous playback controls throughout all application screens.

[6] Calorie & Macro Calculator: The Calorie Calculator accepts user inputs for age, height (cm), weight (kg), number of daily meals, gender, and fitness goal. Upon submission, the module applies the Harris-Benedict BMR formula and computes a personalised TDEE alongside macro targets for protein, fat, and carbohydrates. When multiple meals are specified, the results view presents an additional horizontally scrollable per-meal breakdown, allowing users to plan individual meal compositions. A dedicated results section beneath the macro display surfaces intelligently matched recipe suggestions whose macro profiles fall within the acceptable error threshold.

5. Conclusion:

We are pleased to announce the successful development and deployment of the Fit&Fuel application, in alignment with the initial project objectives. The development team rigorously adhered to the project's specifications and implemented a thorough testing regimen to ensure a robust and reliable product for fitness enthusiasts. Fit&Fuel represents a significant advancement in addressing the fragmentation and connectivity constraints prevalent in the mobile fitness application market.

By providing users with a unified, offline-first platform that integrates exercise guidance, personalised nutrition calculation, recipe management, and motivational audio, Fit&Fuel empowers individuals to pursue comprehensive health management without dependence on external services or network availability. The application's modular Flutter architecture, powered by GetX's reactive state management, ensures that the codebase remains maintainable and extensible as new capabilities are introduced.

Moving forward, the team is committed to further enhancing the system by incorporating user authentication and cloud synchronisation for cross-device data persistence, wearable device integration for real-time biometric feedback, adaptive workout programming based on progressive overload principles, expanded recipe databases with allergen filtering, and AI-driven personalisation of exercise and nutrition recommendations. These enhancements will position Fit&Fuel as a comprehensive, intelligent health companion capable of serving the evolving needs of the fitness community.

6. Reference:

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